

Natasha Sieh

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BUILDING @ University of Michigan, Ann Arbor

RL Control Engineer & Subteam Lead | *Michigan Wolverbot Kickers (RoboCup)* | Jan 2026 – Present

- **URDF Reconstruction:** Diagnosed and rebuilt the team's broken humanoid URDF from scratch — re-authored all joint links and component connections to achieve a stable simulation model. Validating dynamics in MuJoCo and training initial ball-contact policies in Isaac Lab.
- **Fall Recovery:** Implementing RL models in Isaac Lab to optimize humanoid fall-recovery trajectories across diverse orientations; developing evaluation frameworks for velocity vector magnitude and postural stability.
- **Sim-to-Real:** Engineering transfer logic to bridge Isaac Lab physics and physical hardware — closing the gap between learned policy and real legged locomotion dynamics.
- **RoboCup 2026:** Attending RoboCup Humanoid League (Incheon, Korea) to scout top teams for next season.

Navigation Engineer | *Michigan Autonomous Robot Vehicle (UMARV)* | Sept 2024 – April 2025

- **Motion Planning:** Refined A* and BFS search heuristics in C++ to reduce node expansions — 50% faster motion planning and best UMARV performance at IGVC 2025. Co-authored IGVC Design Report.

RESEARCH @ University of Michigan, Ann Arbor

Undergraduate Researcher | *Field Robotics Group — Prof. Skinner* | May 2025 – April 2026

- **Sonar Simulator:** Built a GPU-accelerated side-scan sonar simulator in NVIDIA Isaac Sim using CUDA, PhysX, and Warp to model acoustic beam patterns and ray-tracing to replicate underwater acoustic physics without costly at-sea testing. Validated against real-world shipwreck sonar imagery. Co-authored publication submitted to OCEANS 2026.
- **Terrain Synthesis:** Engineered a Synthetic Data Generation pipeline using PyTorch, VGG-19 Neural Style Transfer, and parametric Perlin noise to simulate natural seafloor textures learned from real sonar datasets, bridging the sim-to-real gap for underwater perception.
- **Infrastructure:** Implemented ROS-to-ROS2 node bridge for simulation interoperability across the lab's pipeline.

Undergraduate Researcher | *Extracorporeal Life Support Lab — Prof. Potkay* | Sept 2024 – April 2025

- **Hardware-Software Integration:** Implemented C++ embedded temperature control (thermistors + PID feedback) for wearable artificial lung prototypes, doubling fabricated thickness with fewer defects.

INDUSTRY @ Naval Information Warfare Center (NIWC) Pacific

RF/Autonomous Systems Intern (SEAP) | San Diego, CA | June 2024 – Aug 2024

- **Automation:** Built a Python-driven GUI to automate RF-over-fiber calibration with real-time SFDR and Noise Floor algorithms, reducing human error and data collection time 40%.

Autonomous Navigation Intern (SEAP) | Honolulu, HI | June 2023 – Aug 2023

- **Quadruped Navigation:** Integrated LiDAR and RGB-D for multi-modal mapping; OpenCV object recognition pipeline; Gazebo/ROS 2D SLAM for real-time localization and path planning in unstructured environments.

SELECTED PROJECTS @ University of Michigan, Ann Arbor

- **Autonomous Swarm for Ecological Restoration:** Decentralized swarm stack for autonomous seagrass site identification in low-bandwidth underwater environments; benchmarked 3 algorithms against RMSE & confusion matrices.
- **Autonomous Robot Stack:** Custom multithreaded pub-sub middleware, FK/IK kinematics, A* motion planning;

SKILLS

Simulation & GPU: Isaac Sim/Lab, CUDA, PhysX, Warp, MuJoCo, Gazebo

Perception & ML: PyTorch, VGG-19, OpenCV, SLAM, SDG, Sensor Fusion

Planning & Control: A*/BFS, RL, AFSA, FK/IK, PID, ROS/ROS2

Languages & Embedded: C/C++, Python, Julia, Linux, Git · ESP32/Arduino/RPi

RECOGNITIONS

CRA Outstanding Undergraduate Researcher Honorable Mention (2026) — National

UROP Summer Research Fellowship (2025), UMich

IGVC Design Report National Finalist (2025)

Course Grader, Computational Linear Algebra (W2025)

FIRST Robotics Team 2438 — IMPACT & Chairman's Award, 2023 & 2024

EDUCATION

University of Michigan, Ann Arbor, B.S. Robotics Engineering, expected May 2028. GPA 3.66. Coursework includes: Dynamics and Vibrations, ROS, Marine Robotics.